Arista Access Node KR-ACC-82 SOP for Minor Optical Transport Failure (SFP/XFP degradation, fiber cuts, Rx/Tx power loss)

Node: KR-ACC-82 | **Vendor:** Arista | **Severity:** Minor | **Alarm Code:** ALARM_6363

Description

This Standard Operating Procedure (SOP) outlines the steps required to diagnose and resolve a minor optical transport failure on the Arista Access Node KR-ACC-82 at the KR Puram Access site. The failure domain of interest is related to SFP/XFP degradation, fiber cuts, or Rx/Tx power loss, which can lead to a loss of connectivity between the access node and the core network. This failure domain is specific to the Arista Access Node due to its reliance on optical transport modules (SFP/XFP) to transmit and receive data over optical fibers. The Arista Access Node's hardware and software design utilizes the Arista Extensible Operating System (EOS) to manage the optical transport modules and ensure reliable data transmission. However, hardware degradation or physical damage to the SFP/XFP or the optical fiber can cause a loss of signal strength, leading to a minor optical transport failure. This SOP is designed to help network operators quickly diagnose and resolve such failures to minimize downtime and ensure continuous network availability.

Pre-requisite Checks

- pre_requisite: Verify that the KR Puram Access site is accessible and the remote hands are present to assist in the troubleshooting process., condition: Ensure that the redundant link status is checked and confirmed to be operational.

- pre_requisite: Verify that the network operations team has been notified of the failure and is aware of the troubleshooting process., condition: Ensure that the necessary spare parts (SFP/XFP or fiber) are available at the site.

Potential Root Causes

Cause	Probability
SFP/XFP degradation	High
Fiber cuts or physical damage	High
Rx/Tx power loss due to hardware or software issues	Medium

Diagnostic Commands & Expected Outputs

```
Command: show interface 1/1/1, Expected Output: Interface 1/1/1, Ethernet, 100GigE, up Hardware: 100GigE, SFP-100G-CWDM4 Description: KR-Puram-Access-Link-1 MTU: 9216 Speed: 100Gbit Duplex: Full Auto-negotiation: on Link length: 10km SFP status: present SFP type: CWDM 4x100G SFP rate: 100Gbit, Description: This command is used to gather information about the interface 1/1/1, including the SFP status
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and type, which can help identify if the issue is related to the SFP or the fiber.

Command: show environment, Expected Output: Environment Information: Temperature: 22C Humidity: 60% Power supply 1: present Power supply 2: present, Description: This command is used to gather information about the environment, including temperature, humidity, and power supply status, which can help identify if the issue is related to environmental factors.
```

Resolution Steps

Step 1: Step: Replace the SFP/XFP module, Command: replace module 1/1/1, Description: This step involves replacing the SFP/XFP module on interface 1/1/1 with a new one. Make sure to handle the old module carefully and store it in a safe place for further analysis.

Step 2: Step: Verify the fiber connection, Command: show interface 1/1/1, Description: This step involves verifying the fiber connection to ensure that it is securely connected to the SFP/XFP module and the other end of the fiber is properly connected to the core network.

Post-Check Verification

```
Command: show interface 1/1/1, Expected Output: Interface 1/1/1, Ethernet, 100GigE, up Hardware: 100GigE, SFP-100G-CWDM4 Description: KR-Puram-Access-Link-1 MTU: 9216 Speed: 100Gbit Duplex: Full Auto-negotiation: on Link length: 10km SFP status: present SFP type: CWDM 4x100G SFP rate: 100Gbit, Description: This command is used to verify that the interface 1/1/1 is up and running properly.
```

Roll-back Procedures

- Step: Restore the original SFP/XFP module, Command: restore module 1/1/1, Description: This step involves restoring the original SFP/XFP module that was replaced. Make sure to handle the new module carefully and store it in a safe place for further analysis.